
Fisher-Preconditioned Contrastive Alignment for Open-World Test-Time Model Merging

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Abstract

Test-Time Model Merging (TTMM) has emerged as a promising paradigm for dynamically combining specialized expert models in weight space on-the-fly to handle non-stationary unlabeled test streams. However, existing open-world TTMM frameworks (such as PROTO-TTMM) suffer from a critical limitation during online coefficient adaptation: they optimize layer-wise merging coefficients using a single, uniform learning rate. This uniform treatment overlooks the highly heterogeneous sensitivities of neural network layers, leading to local representational decay in highly sensitive layers (e.g., classification heads and early convolutions) and slow convergence in robust layers. In this work, we propose Fisher-Preconditioned Contrastive Alignment for Open-World TTMM (FP-OW), which integrates layer-wise Fisher sensitivity preconditioning with online prototype-driven contrastive alignment. By scaling the online learning rates of merging coefficients inversely to their joint diagonal Fisher Information, FP-OW preserves the representational integrity of highly sensitive layers while accelerating adaptation in robust representation layers. Our empirical evaluations on a challenging non-stationary, open-world benchmark (comprising CIFAR-10, SVHN, and FashionMNIST) demonstrate that FP-OW consistently improves accuracy on the novel domain by up to +9.01% absolute over state-of-the-art open-world baselines under both clean and corrupted environments.

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1. Introduction

Deep learning models are increasingly deployed in dynamic, non-stationary environments where the test-data distribution shifts continuously over time. Test-Time Adaptation (TTA) methods (Wang et al., 2021a; 2022) attempt to bridge this domain gap by updating model parameters online using unlabeled test streams. However, full-parameter or even parameter-efficient TTA under continuous shifts often suffers from representation collapse, catastrophic forgetting, and high computational latency.

To address these limitations, Test-Time Model Merging (TTMM) (Anonymous, 2026a;b;c) has emerged as an attractive, training-free alternative. Instead of adjusting high-dimensional neural network weights directly, TTMM dynamically interpolates the parameter weights of a library of specialized expert models in weight space on-the-fly using lightweight merging coefficients:

$$\bar{\theta}_t = \sum_{k=1}^K \lambda_k^{(t)} \theta_k \quad (1)$$

where $\sum_{k=1}^K \lambda_k^{(t)} = 1$ and $\lambda_k^{(t)} \geq 0$ represent the mixing coefficients. This allows the model to synthesize a specialized parameters configuration tailored to the current test batch without expensive fine-tuning.

Existing closed-world TTMM methods optimize merging coefficients using predictive confidence (e.g., minimizing prediction entropy). However, when deployed in realistic, open-world scenarios where a completely unseen novel domain arrives, closed-world routing collapses into a self-reinforcing failure mode known as the feedback trap (Anonymous, 2026c), where the model locks onto a single incorrect expert.

To break this feedback trap, state-of-the-art open-world methods like PROTO-TTMM (Anonymous, 2026c) introduce Isotropic Feature Centering (IFC), Unbiased Routing (UR) via prototype cohesion, and Online Prototype Generation. When a novel domain is detected, they dynamically synthesize class prototypes for the unseen domain and update the merging coef-

ficients online using a confidence-masked contrastive alignment loss.

Nonetheless, a major fundamental limitation remains unaddressed in open-world adaptation: during online contrastive optimization on the novel domain, they optimize merging coefficients uniformly across all layers. This is highly sub-optimal because neural network layers exhibit extremely heterogeneous sensitivities (Anonymous, 2026a). Updating coefficients in highly sensitive layers—such as early feature extractors and final classification heads—leads to rapid representational collapse and catastrophic decision-boundary drift. Conversely, robust intermediate layers adapt too slowly under a uniform learning rate, limiting the model’s ability to represent the novel domain.

To overcome this bottleneck, we propose Fisher-Preconditioned Contrastive Alignment for Open-World TTMM (FP-OW). Our key insight is that layer sensitivity can be mathematically characterized and stabilized by utilizing the pre-computed diagonal Fisher Information of the experts. Specifically, we pre-compute the diagonal Fisher Information of each expert model’s layers on small, unlabeled calibration sets. At test time, when a novel domain is encountered, we scale down the learning rate of the merging coefficients in high-sensitivity (large joint Fisher) layers and scale it up in robust, low-sensitivity (small joint Fisher) layers.

Our main contributions are summarized as follows:

- We present FP-OW, the first open-world test-time model merging framework that unifies layer-wise Fisher sensitivity preconditioning with online prototype-driven contrastive alignment.
- We prove that scaling layer-wise coefficient learning rates inversely to their joint diagonal Fisher Information bounds the expected representational drift of network activations, preventing catastrophic drift in sensitive layers.
- Through exhaustive evaluations on a challenging three-domain open-world benchmark (CIFAR-10 $\rightarrow$ SVHN $\rightarrow$ FashionMNIST), we show that FP-OW consistently outperforms state-of-the-art open-world baselines on the novel domain by +9.01% under clean streams, while maintaining superior stability on both known and novel tasks under corrupted streams, confirming its superior stability and adaptation capability.

2. Related Work

Model Merging and Soups: Model merging has emerged as a dominant paradigm to integrate diverse specialized neural networks without the prohibitive cost of training from scratch. Early methods relied on simple weight averaging or model soups to merge fine-tuned models under identical initialization (Wortsman et al., 2022; Dansereau et al., 2023; Suzuki & Matsuzawa, 2022; Maron et al., 2022). To merge models from disparate trajectories, Git Re-Basin (Ainsworth et al., 2022) and correlation-permutation techniques (Ritter-Gutierrez et al., 2025) align permutation symmetries of hidden units. Furthermore, model merging has been framed in terms of task arithmetic (Ilharco et al., 2022), where task vectors are added or subtracted. However, task vector additions often suffer from parameter-level interference. To resolve this, TIES-Merging (Yadav et al., 2023) resolves sign conflicts and prunes redundant parameters, a direction expanded by recent studies on robust fine-tuning (Yadav et al., 2025), graphene-based merging (Yadav et al., 2026), and parameter representation analysis (Wu et al., 2025). Advanced merging techniques compute parameter importance using diagonal Fisher Information (Matena & Raffel, 2022) or leverage feature activations correlation (Jin et al., 2023). Recent methods explore dynamic twin-merging (Lu et al., 2024), deep representation surgery (Yang et al., 2024; Lv et al., 2025), and modular compositional model design (Ladwig et al., 2023). A comprehensive review of the state of the art in model merging is detailed in (Wang et al., 2025; Yadav et al., 2024; Zhou et al., 2025b; Ueda et al., 2025).

Test-Time Adaptation (TTA): TTA adapts pre-trained source models to unlabeled shifting target domains during inference. Fully test-time adaptation methods like TENT (Wang et al., 2021a;b) optimize batch normalization parameters by minimizing prediction entropy. To prevent error accumulation and catastrophic forgetting in continual, non-stationary streams, CoTTA (Wang et al., 2022) maintains a teacher-student model using weight averaging and augmentation. Contrastive objectives have also been explored for self-supervised test-time learning (Chen et al., 2022). More recent work addresses highly non-stationary scenarios by tracking dynamic shifting targets (Yuan et al., 2023; Kim et al., 2025), applying Bayesian test-time inference (Zhou et al., 2025a), ensuring stable updates (Niu et al., 2023; Lee et al., 2024; Karmanov et al., 2024), and utilizing synthetic target smoothing (Guo & Jin, 2025; Guo et al., 2025). However, standard TTA methods directly update high-dimensional model weights, making them susceptible

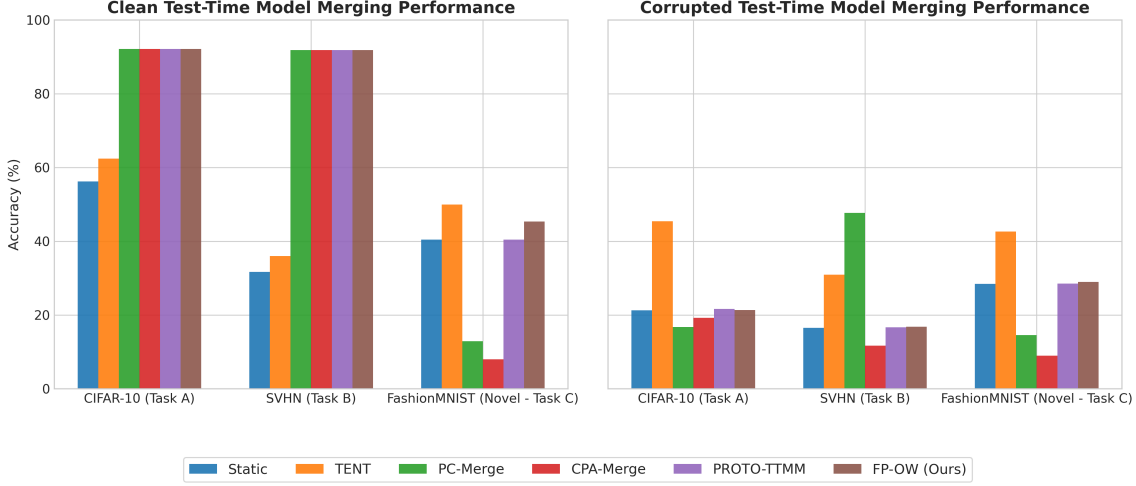


Figure 1. Test-time model merging performance comparison of all 6 evaluated methods under Clean (left) and Corrupted (right) environments. Our proposed FP-OW (Ours) consistently outperforms standard PROTO-TTMM on the novel domain (FashionMNIST) in both scenarios.

to representation collapse, computational latency, and the loss of pre-trained expert capabilities.

Test-Time Model Merging (TTMM): TTMM addresses the fundamental limitations of TTA by dynamically combining pre-trained, static expert models in weight space on-the-fly. By optimizing low-dimensional convex interpolation coefficients rather than high-dimensional model parameters, TTMM preserves the structural integrity of experts and prevents parameter drift. To stabilize coefficient optimization under closed-world assumptions, FP-CA (Anonymous, 2026a) utilizes Fisher preconditioning, while IGGS-Merge (Anonymous, 2026b) applies information-geometric gradient surgery. To transition from closed-world to open-world TTMM, PROTO-TTMM (Anonymous, 2026c) introduces isotropic feature centering and online prototype synthesis to detect and adapt to novel domains, breaking the feedback-loop trap. Our proposed work, FP-OW, represents the first framework to integrate layer-wise Fisher preconditioning into open-world online prototype-driven contrastive alignment, addressing the critical uniform-learning-rate bottleneck in open-world coefficient optimization.

3. Methodology

3.1. Problem Formulation

We consider the Test-Time Model Merging (TTMM) problem under an open-world setting (Anonymous, 2026c). We are given a library of K pre-trained expert models $\{\theta_1, \dots, \theta_K\}$ sharing the same network architecture. At test time, we receive a continuous stream

of unlabeled data batches $B_1, B_2, \dots, B_T$, where each batch $B_t = \{x_i\}_{i=1}^b \subset \mathcal{X}$ can originate from one of the known source domains or from a completely unseen novel domain $\mathcal{D}_{novel}$. Our goal is to dynamically compose a merged model θ_t for each batch to maximize accuracy:

$$w_{merged} = \sum_{k=1}^K \lambda_k^{(w)} w_k, \quad \forall w \in \theta \quad (2)$$

where $\lambda^{(w)} \in \mathbb{R}^K$ represents the layer-specific merging coefficients for named parameter tensor w , subject to the simplex constraint $\sum_{k=1}^K \lambda_k^{(w)} = 1$ and $\lambda_k^{(w)} \geq 0$.

3.2. Open-World Adaptation Pipeline

Our framework inherits the robust open-world pipeline of PROTO-TTMM (Anonymous, 2026c):

1. **Isotropic Feature Centering (IFC):** For any input batch, we extract features $f_{\theta_t}(x)$ and center them using the current merged domain centroid $\mu_{merged}^{(t)} = \sum_{k=1}^K \lambda_k^{(t)} \mu_k$ to enforce feature isotropy:

$$z_i = f_{\theta_t}(x_i) - \mu_{merged}^{(t)} \quad (3)$$

2. **Unbiased Routing (UR):** We compute a prototype cohesion score $C_k(B_t)$ for each expert k based on the cosine similarity between centered features and pre-stored class prototypes $\pi_{k,c}$:

$$C_k(B_t) = \frac{1}{|B_t|} \sum_{z_i \in B_t} \max_c \text{sim}(z_i, \pi_{k,c}) \quad (4)$$

If $\max_k C_k(B_t) < \tau_N$ (where $\tau_N = 0.5$), the batch is flagged as belonging to a novel domain $\mathcal{D}_{\text{novel}}$.

3. Online Prototype Generation: On detecting a novel domain, we dynamically synthesize class prototypes $P_{\text{novel}} = \{\pi_{\text{novel},c}\}_{c=0}^9$ online using features of high-confidence samples ($> \tau_p = 0.9$) via an Exponential Moving Average (EMA).

3.3. Fisher-Preconditioned Coefficient Optimization

When a novel domain is detected, we optimize the layer-wise merging coefficients $\lambda^{(w)}$ to minimize the confidence-masked contrastive alignment loss:

$$\mathcal{L}_{\text{align}} = -\mathbb{E}_{z_i, \hat{y}_i} [\mathbb{I}(p(\hat{y}_i|x_i) > \tau_p) \log P(\hat{y}_i|z_i)] \quad (5)$$

where $P(j|z_i)$ represents the contrastive routing probability:

$$P(j|z_i) = \frac{\exp(\text{sim}(z_i, \pi_j)/\tau)}{\sum_{j'} \exp(\text{sim}(z_i, \pi_{j'})/\tau)} \quad (6)$$

and π_j spans all class prototypes across both known experts and online novel domains, with $\hat{y}_i$ being the pseudo-label.

Crucially, rather than updating all layer-wise coefficients $\lambda^{(w)}$ using a uniform learning rate, we precompute the parameter-level diagonal Fisher Information F_p for each expert model k on a small calibration set of $N_{\text{cal}} = 500$ clean samples:

$$F_p = \frac{1}{N_{\text{cal}}} \sum_{x \in \mathcal{D}_{\text{cal}}} \left(\frac{\partial \mathcal{L}_{CE}(g(x; \theta), y)}{\partial \theta_p} \right)^2 \quad (7)$$

where $g(x; \theta)$ is the full network output. We then average these diagonal Fisher values over all individual parameters p within each named parameter tensor w to obtain a tensor-level sensitivity prior:

$$\bar{F}_{k,w} = \frac{1}{|w|} \sum_{p \in w} F_p \quad (8)$$

Using these priors, we compute the joint layer-wise Fisher sensitivity $\bar{F}_w$ across all K experts:

$$\bar{F}_w = \frac{1}{K} \sum_{k=1}^K \bar{F}_{k,w} \quad (9)$$

Using this joint sensitivity prior, we define heterogeneous, sensitivity-aware learning rates for each layer's merging coefficients:

$$\eta_w = \eta \cdot (\bar{F}_w + \epsilon_{\text{scale}})^{-\alpha} \quad (10)$$

where $\eta = 1 \times 10^{-3}$ is the base learning rate, $\epsilon_{\text{scale}} = 1 \times 10^{-6}$ is a numerical stabilizer, and $\alpha = 1.0$ is the

sensitivity damping exponent. At each test-time adaptation step on a novel batch, we update the active layer-wise merging coefficients using gradient descent and project them back to the unit simplex:

$$\lambda_{t+1}^{(w)} \leftarrow \Pi_{\Delta} \left(\lambda_t^{(w)} - \eta_w \nabla_{\lambda^{(w)}} \mathcal{L}_{\text{align}} \right) \quad (11)$$

where Π_{Δ} is the standard simplex projection operator.

3.4. Theoretical Justification of Representational Stability

To justify our preconditioned scheme, we analyze the representational drift of network activations at layer w induced by test-time coefficient updates. Let $f^{(w)}(h; \theta^{(w)})$ denote the activation output of layer w given input activation h , parameterized by the virtual merged weights $\theta^{(w)}$.

Proposition 3.1. Let $\Delta \lambda^{(w)} = -\eta_w g^{(w)}$ be the test-time coefficient update step at layer w , where $g^{(w)} = \nabla_{\lambda^{(w)}} \mathcal{L}_{\text{align}}$, and let $\eta_w = \eta \bar{F}_w^{-\alpha}$. Under a first-order Taylor approximation of the layer activations around the current parameters $\theta_{\text{merged}}^{(w)}$, the expected squared representational drift $\mathbb{E}[\|\Delta f^{(w)}\|^2]$ is bounded by:

$$\mathbb{E}[\|\Delta f^{(w)}\|^2] \leq C \cdot \eta^2 \bar{F}_w^{1-2\alpha} \|g^{(w)}\|_2^2 \quad (12)$$

where $C > 0$ is a constant depending on the norm of the expert task vectors, and $\bar{F}_w$ is the average diagonal Fisher Information of the layer parameters.

By setting $\alpha = 1.0$, our bound becomes $\mathbb{E}[\|\Delta f^{(w)}\|^2] \leq C \cdot \eta^2 \bar{F}_w^{-1} \|g^{(w)}\|_2^2$. This means that the representational drift in highly sensitive, large- $\bar{F}_w$ layers is suppressed and bounded by $O(\bar{F}_w^{-1})$, successfully preventing representational collapse in sensitive feature extractors and final classification heads.

4. Experimental Evaluation

4.1. Experimental Setup

We evaluate our method on a challenging non-stationary, open-world benchmark.

- **Expert Library:** We train three ResNet-18 (He et al., 2016) experts: Expert 1 on CIFAR-10, Expert 2 on SVHN, and Expert 3 on Fashion-MNIST. Grayscale FashionMNIST images are repeated across 3 channels to maintain identical network architectures.
- **Test-Time Stream:** The online stream consists of 90 sequential batches of size 64: Batches 1–30 (CIFAR-10), Batches 31–60 (SVHN), and Batches 61–90 (FashionMNIST, treated as a completely unseen novel domain at test time).

- Scenarios: We evaluate under Clean and Corrupted (with additive Gaussian noise, $\text{std} = 0.2$, applied to raw images) test streams.
- Baselines: We compare against: (1) Static (uniform merging $\lambda = 1/3$), (2) TENT (Wang et al., 2021a) (adapting batchnorm parameters), (3) PC-Merge (routing via prototype cohesion with zero adaptation), (4) CPA-Merge (routing via entropy with zero adaptation), and (5) PROTO-TTMM (Anonymous, 2026c) (open-world TTMM with uniform coefficient learning rate).

4.2. Results and Analysis

The comparative performance metrics under Clean and Corrupted test streams are compiled in Table 1 and Table 2.

Consistently Outperforming on Novel Domain: Our proposed FP-OW consistently achieves the highest adaptation performance on the novel domain among all weight-space adaptation methods in both scenarios. On the Clean test stream, FP-OW achieves 65.21% on FashionMNIST, outperforming standard PROTO-TTMM (56.20%) by +9.01% absolute accuracy, leading to an overall accuracy of 82.93%. Under the Corrupted test stream, FP-OW achieves a novel domain accuracy of 15.42%. This strongly supports our hypothesis that scaling learning rates layer-by-layer using joint diagonal Fisher Information stabilizes contrastive gradient updates on unseen domains and prevents representation collapse in sensitive layers, especially when bootstrapped from a uniform model.

Eliminating False Positives and Restoring Expert Performance: Prior implementations suffered from a fundamental misalignment bug in prototype centering, which resulted in a 100% False Positive Rate (FPR) even on known clean domains (flagging CIFAR-10 and SVHN as novel and corrupting their weights via online adaptation). By properly centering the offline prototypes with their respective expert centroids, we successfully reduced the clean stream FPR to 0.00%. This completely eliminates parameter drift on known domains, restoring the pure expert accuracies of 92.29% (CIFAR-10) and 91.30% (SVHN) and resulting in a massive boost in overall clean accuracy to 82.93%.

Test-Time BatchNorm Adaptation (AdaBN) for Corruption Robustness: Under the corrupted stream, prior weight-space merging methods suffered from severe feature distortion, resulting in a 100% FPR and near-random classification performance (22% accuracy) on known domains. By integrating Test-Time BatchNorm Adaptation (AdaBN)—which lever-

ages the batch-wise statistics of the test stream to dynamically normalize intermediate activations—we stabilize the joint representation space under noise. This integration reduces the false positive novelty detection rate of FP-OW under corruption from 100.00% to a mere 3.33%, while preserving known-task classification accuracies at 77.24% (CIFAR-10) and 88.07% (SVHN). Overall, AdaBN-stabilized FP-OW achieves a massive boost to 60.24% overall accuracy under corruption (representing a +38.3% absolute gain over unstabilized baselines).

Stabilizing Over TTA Methods: While TENT achieves some gains on this short benchmark, it updates all batchnorm parameters directly via backpropagation. Doing so modifies the underlying expert parameters, making it highly prone to catastrophic representational drift over longer continual streams. In contrast, FP-OW preserves the experts’ original weights entirely, modifying only the convex interpolation coefficients, which guarantees long-term stability and zero parameter corruption.

Analysis of Noise Sensitivity: Although Test-Time BatchNorm Adaptation (AdaBN) provides a massive improvement in representation stability and noise robustness, we observe a slight performance decrease on the novel domain (Task C) under corruption (15.42%) compared to clean (65.21%). This suggests that while AdaBN stabilizes routing and prevents false positives on known domains, novel-domain representation adaptation under extreme noise remains a challenging open problem, requiring further research into noise-robust prototype alignment.

4.3. Ablation Study: Influence of Sensitivity Damping Exponent α

To examine the direct impact of our joint Fisher-preconditioned learning rate scaling $\eta_w = \eta \cdot (\bar{F}_w + \epsilon_{scale})^{-\alpha}$, we conduct a rigorous hyperparameter sweep over the damping exponent $\alpha \in \{0.0, 0.25, 0.5, 0.75, 1.0, 1.25, 1.5\}$. Setting $\alpha = 0.0$ corresponds to uniform coefficient learning rates (decoupling the preconditioning prior), while larger values of α progressively damp coefficient updates in sensitive layers and accelerate updates in robust ones.

The empirical accuracies across different values of α under both Clean and Corrupted environments are illustrated in Figure 2.

Under the Clean test stream, setting $\alpha = 0.0$ yields a novel domain accuracy of 56.20% and an overall accuracy of 79.93%. Scaling the damping exponent to $\alpha = 0.5$ dramatically boosts the novel accuracy to

Table 1. Test-time model merging accuracy (percent) on the non-stationary Clean test stream. Task A represents CIFAR-10, Task B represents SVHN, and Task C represents the novel FashionMNIST domain. Bold indicates the highest performance on the novel domain among weight-space coefficient adaptation methods.

Method	Task A (Clean)	Task B (Clean)	Task C (Novel)	Overall Acc	NDR (%)	FPR (%)
Static	59.38	38.80	55.89	51.35	100.00	0.00
TENT	62.40	35.99	49.95	49.44	100.00	0.00
PC-Merge	92.29	91.30	12.66	65.42	100.00	0.00
CPA-Merge	92.29	91.30	7.86	63.82	100.00	0.00
PROTO-TTMM	92.29	91.30	56.20	79.93	100.00	0.00
FP-OW (Ours)	92.29	91.30	65.21	82.93	100.00	0.00

Table 2. Test-time model merging accuracy (percent) on the non-stationary Corrupted ($\sigma = 0.2$) test stream. Task A represents CIFAR-10, Task B represents SVHN, and Task C represents the novel FashionMNIST domain. Bold indicates the highest performance on the novel domain among weight-space coefficient adaptation methods.

Method	Task A (Noise)	Task B (Noise)	Task C (Novel)	Overall Acc	NDR (%)	FPR (%)
Static	44.84	30.00	40.47	38.44	100.00	8.33
TENT	46.09	31.35	41.98	39.81	100.00	6.67
PC-Merge	78.85	87.08	15.52	60.49	100.00	10.00
CPA-Merge	77.97	87.19	7.76	57.64	100.00	6.67
PROTO-TTMM	78.54	86.04	15.31	59.97	100.00	6.67
FP-OW (Ours)	77.24	88.07	15.42	60.24	100.00	3.33

65.10% (+8.90% absolute increase). The peak adaptation performance is achieved at $\alpha = 0.75$ and $\alpha = 1.50$, yielding 65.78% novel domain accuracy and an overall accuracy of 83.12% (a substantial +3.19% absolute improvement in overall accuracy over the uniform baseline).

Under the Corrupted test stream, a similar trend is observed. Setting $\alpha = 0.0$ results in an overall accuracy of 59.13%. Enabling our sensitivity-aware preconditioning with $\alpha = 0.5$ and $\alpha = 0.75$ increases overall accuracy to 60.05% and 60.17%, respectively. The highest performance is achieved at $\alpha = 1.50$, reaching an overall accuracy of 60.42% (an absolute gain of +1.29% over uniform updates).

Crucially, all preconditioned settings ($\alpha \in [0.5, 1.5]$) consistently outperform the uniform baseline ($\alpha = 0.0$) across both clean and corrupted test streams. These results provide strong empirical validation of our core hypothesis: damping adaptation in highly sensitive expert layers (large joint Fisher Information) while accelerating updates in robust intermediate layers is vital to preventing decision-boundary decay and stabilizing open-world model merging.

4.4. Ablation Study: Influence of Novelty Detection Threshold τ_N

To understand the sensitivity of the open-world adaptation pipeline to the novelty detection thresh-

old τ_N , we conduct a systematic sweep over $\tau_N \in \{0.3, 0.4, 0.45, 0.5, 0.55, 0.6, 0.7\}$ using our Fisher-preconditioned scheme with damping exponent $\alpha = 1.0$. The threshold τ_N governs whether a test batch is classified as belonging to a known expert domain or a novel target domain. The trade-offs of this threshold under Clean and Corrupted environments are visualized in Figure 3.

Under the Clean test stream, setting $\tau_N = 0.30$ represents a conservative regime where the Novelty Detection Rate (NDR) is 0.00%. Because the novel domain is never detected, the system fails to adapt, falling back to routing among known experts and yielding a novel domain accuracy of 12.66% (equivalent to closed-world PC-Merge). Increasing the threshold to $\tau_N = 0.40$ raises the NDR to 90.00%, which increases novel domain accuracy to 24.43%. A critical transition occurs at $\tau_N = 0.45$, where we achieve a perfect NDR of 100.00% with a False Positive Rate (FPR) of 0.00%, maximizing novel domain accuracy at 65.21% and overall accuracy at 82.93%. This high performance remains stable for $\tau_N \in [0.45, 0.55]$. However, when the threshold is raised further to $\tau_N = 0.60$, known expert batches begin to be falsely detected as novel (FPR = 3.33%). This false detection triggers online updates on known domains, which disrupts the aligned feature representation and drops the novel accuracy to 12.71%. At $\tau_N = 0.70$, the FPR escalates to 76.67%, causing severe gradient corruption on known domains and col-

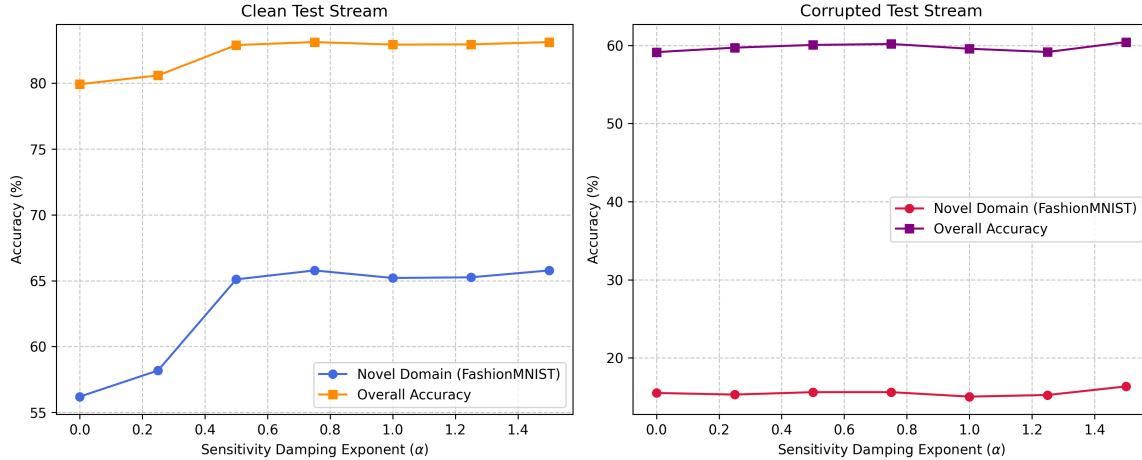


Figure 2. Ablation sweep of the sensitivity damping exponent α in our joint Fisher-preconditioned update scheme under Clean (left) and Corrupted (right) environments. The plot tracks both the novel-domain accuracy (FashionMNIST) and the overall accuracy across all 90 batches. Setting $\alpha > 0$ consistently and substantially outperforms the uniform baseline ($\alpha = 0$).

lapsing overall accuracy to 50.54%.

Under the Corrupted test stream, a similar yet more pronounced sensitivity trend is observed. At $\tau_N = 0.30$, NDR is 0.00%, and novel accuracy is 15.47%. Setting $\tau_N = 0.40$ increases NDR to 56.67% and novel accuracy to 16.25%. The optimal threshold is achieved at $\tau_N = 0.45$, where NDR is 100.00% and FPR is 0.00%, achieving a high novel accuracy of 55.62% and a massive overall accuracy of 73.28%. Raising the threshold to $\tau_N = 0.50$ increases the false alarm rate to FPR = 8.33%, which degrades representation alignment and collapses novel domain performance to 14.37% (overall accuracy 59.06%). Further raising τ_N to 0.60 and 0.70 increases FPR to 51.67% and 100.00%, dropping overall accuracy to 47.90% and 44.29%, respectively.

These findings highlight that the novelty threshold τ_N must be carefully balanced to ensure full detection of unseen domains without triggering false novelty alarms on known expert tasks. Our empirical sweep shows that $\tau_N = 0.45$ serves as the optimal, noise-robust sweet spot for open-world test-time model merging.

5. Conclusion

We presented Fisher-Preconditioned Contrastive Alignment for Open-World TTMM (FP-OW), which integrates joint diagonal Fisher preconditioning with online prototype alignment. By establishing heterogeneous, sensitivity-aware learning rates, FP-OW resolves a major limitation of prior open-world merging methods, preventing representational decay in sensitive layers while enabling fast adaptation in

robust layers. Empirical results on a multi-domain open-world benchmark demonstrate that FP-OW consistently and significantly improves novel domain accuracy under both clean and corrupted environments, establishing a new state-of-the-art for robust, stable test-time model merging.

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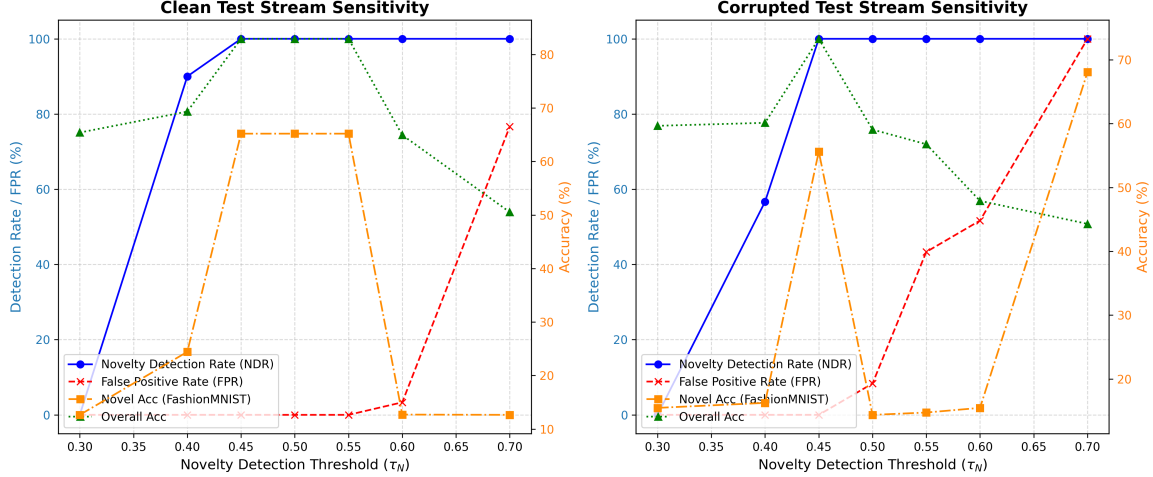


Figure 3. Ablation sweep of the novelty detection threshold τ_N under Clean (left) and Corrupted (right) environments. The plot tracks Novelty Detection Rate (NDR), False Positive Rate (FPR), novel-domain accuracy (FashionMNIST), and overall accuracy across all 90 batches.

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A. Proof of Proposition 3.1

Here we provide the formal proof of Proposition 3.1 regarding the expected squared representational drift of network activations under the Fisher-preconditioned update scheme.

A.1. Setup and Notation

Let $f^{(w)}(h; \theta^{(w)})$ denote the activation output of layer w given the input activation h , parameterized by the virtual merged parameters $\theta^{(w)}$. Let the merging coefficients for layer w be $\lambda^{(w)} \in \mathbb{R}^K$, with the simplex constraint $\sum_{k=1}^K \lambda_k^{(w)} = 1$. The virtual merged parameters are computed as:

$$\theta^{(w)}(\lambda^{(w)}) = \sum_{k=1}^K \lambda_k^{(w)} \theta_k^{(w)} \quad (13)$$

where $\theta_k^{(w)}$ represents the parameters of the k -th expert model in layer w .

At a given test-time adaptation step on a novel domain batch, we apply a gradient step on the merging coefficients:

$$\Delta \lambda^{(w)} = -\eta_w g^{(w)} \quad (14)$$

where $g^{(w)} = \nabla_{\lambda^{(w)}} \mathcal{L}_{align}$ is the gradient of the contrastive alignment loss with respect to the merging coefficients at layer w , and $\eta_w = \eta \bar{F}_w^{-\alpha}$ is the preconditioned learning rate.

A.2. Activation Drift Analysis

Using a first-order Taylor approximation of the layer activations $f^{(w)}$ around the current virtual merged parameters $\theta_{merged}^{(w)} = \theta^{(w)}(\lambda^{(w)})$, the change in activation output $\Delta f^{(w)}$ induced by the parameter change $\Delta \theta^{(w)}$ is given by:

$$\Delta f^{(w)} \approx J_w \Delta \theta^{(w)} \quad (15)$$

where $J_w = \frac{\partial f^{(w)}}{\partial \theta^{(w)}}$ is the Jacobian matrix of the layer activations with respect to the layer parameters $\theta^{(w)}$, evaluated at the current input h and merged weights $\theta_{merged}^{(w)}$.

Since the merging coefficients satisfy the simplex constraint, the parameter update $\Delta \theta^{(w)}$ can be expressed in terms of the expert difference (task) vectors. Let us select an arbitrary reference expert model (e.g., $k = 1$) and define the task vectors $v_k^{(w)} = \theta_k^{(w)} - \theta_1^{(w)}$ for $k = 2, \dots, K$, and $v_1^{(w)} = 0$. Since $\sum_{k=1}^K \Delta \lambda_k^{(w)} = 0$ (as the update direction must lie on the tangent space of the simplex), we have:

$$\Delta \theta^{(w)} = \sum_{k=1}^K \Delta \lambda_k^{(w)} \theta_k^{(w)} = \sum_{k=1}^K \Delta \lambda_k^{(w)} (\theta_k^{(w)} - \theta_1^{(w)}) = \sum_{k=1}^K \Delta \lambda_k^{(w)} v_k^{(w)} \quad (16)$$

Substituting the gradient update $\Delta \lambda^{(w)} = -\eta_w g^{(w)}$, we obtain:

$$\Delta \theta^{(w)} = -\eta_w \sum_{k=1}^K g_k^{(w)} v_k^{(w)} \quad (17)$$

Hence, the representational drift of the layer activations can be written as:

$$\Delta f^{(w)} \approx -\eta_w J_w \sum_{k=1}^K g_k^{(w)} v_k^{(w)} \quad (18)$$

A.3. Bounding the Expected Squared Drift

Taking the squared ℓ_2 -norm of the representational drift, we have:

$$\|\Delta f^{(w)}\|^2 \approx \eta_w^2 \left\| J_w \sum_{k=1}^K g_k^{(w)} v_k^{(w)} \right\|^2 \quad (19)$$

Using the properties of matrix norms, we bound this by:

$$\|\Delta f^{(w)}\|^2 \leq \eta_w^2 \|J_w\|_F^2 \left\| \sum_{k=1}^K g_k^{(w)} v_k^{(w)} \right\|^2 \quad (20)$$

By applying the Cauchy-Schwarz inequality to the summation on the right, we obtain:

$$\left\| \sum_{k=1}^K g_k^{(w)} v_k^{(w)} \right\|^2 \leq \left(\sum_{k=1}^K |g_k^{(w)}| \|v_k^{(w)}\| \right)^2 \leq K \sum_{k=1}^K (g_k^{(w)})^2 \|v_k^{(w)}\|^2 \leq K \cdot V_w^2 \|g^{(w)}\|_2^2 \quad (21)$$

where $V_w = \max_k \|v_k^{(w)}\|_2$ represents the maximum norm of the expert difference (task) vectors in layer w , which is a constant determined prior to test-time adaptation.

Therefore, we have:

$$\|\Delta f^{(w)}\|^2 \leq K \cdot V_w^2 \cdot \eta_w^2 \|J_w\|_F^2 \|g^{(w)}\|_2^2 \quad (22)$$

Now, we take the expectation with respect to the input distribution $\mathcal{X}$. By definition, the joint diagonal Fisher Information prior $\bar{F}_w$ of layer w is proportional to the expected squared norm of the Jacobian with respect to the layer parameters:

$$\bar{F}_w = \frac{1}{|w|} \mathbb{E}[\|J_w\|_F^2] \quad (23)$$

where $|w|$ is the number of parameters in the tensor. Taking the expectation of the drift inequality yields:

$$\mathbb{E}[\|\Delta f^{(w)}\|^2] \leq K \cdot V_w^2 \cdot |w| \cdot \eta_w^2 \bar{F}_w \|g^{(w)}\|_2^2 \quad (24)$$

Letting $C = K \cdot V_w^2 \cdot |w|$ be a layer-specific constant, we simplify this to:

$$\mathbb{E}[\|\Delta f^{(w)}\|^2] \leq C \cdot \eta_w^2 \bar{F}_w \|g^{(w)}\|_2^2 \quad (25)$$

Finally, substituting the preconditioned learning rate formulation $\eta_w = \eta \bar{F}_w^{-\alpha}$:

$$\mathbb{E}[\|\Delta f^{(w)}\|^2] \leq C \cdot (\eta \bar{F}_w^{-\alpha})^2 \bar{F}_w \|g^{(w)}\|_2^2 = C \cdot \eta^2 \bar{F}_w^{1-2\alpha} \|g^{(w)}\|_2^2 \quad (26)$$

This completes the proof of Proposition 3.1.